YouTube-DNN-RecSys — Two-tower recall (Covington et al.) in TF + Spark

github.com/chenxingqiang/YouTube-DNN-RecSys · Python TensorFlow · Scala Spark · 60 GitHub stars

1. Event / feature sources

Watch history
video / search ids

Context
device / geo / time

Continuous feats
age / counts

Candidate items
catalog embeddings

2. Dual implementation stacks

Python / TensorFlow
data2tfrecords.py
models/dnn.py (Estimator)
custom layers / TensorBoard
SavedModel export

Scala / Spark
FeatureBuilder.scala
BaseSparkLocal
PredictUserVector
ItemEmbeddingPredictor

3. Two-tower model

User tower
item-id embeddings
average watch history
concat continuous feats
MLP → user vector u

Matching
dot(u, v_i)
sampled softmax
next-watch objective

Item tower
shared item embedding
table v_i
optional side features

4. Train / serve

Train
TF Estimator
optional PS / workers
checkpoints

Eval
hold-out next-item
TensorBoard curves

Online recall
ANN / softmax top-k
user vec × item table

Artifacts
TFRecords
SavedModel / Spark vecs

History embeddings are mean-pooled into a user vector and scored against the item table — the paper's candidate-generation stage.